

Paper Modell - Charging Dock

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Figure. Hero picture.

Overview

This project documents the full process of recreating my everyday charger purely from paper materials. Starting from measuring the original object (Fig.1–2), I explored how different paper types behave (Fig.3) and built the internal structure using layered stocks (Fig.4–9). I used the laser cutter to create the logo and the charging-part details (Fig.10–14), then assembled and refined the

final form with attention to curves and fitting (Fig.15–18). The final model (Fig.19–20) aims to stay as close as possible to the real dimensions and geometry.

Reflection

I had never built a paper-based model before, so understanding how each material behaves was a real learning curve. Especially the rounded parts, which simply don’t act like the real charger’s geometry. The internal stick-structure was the hardest part, but it also taught me the most about precision, layering, and maintaining stability.

Using the laser cutter for both the logo and the charging section pushed me to think about how different fabrication techniques support the final quality. Most importantly, I finally felt the difference between imagining an object and actually constructing it: measuring, adjusting, fixing, failing, and retrying. It was stressful, but I’m genuinely proud of how close the model came to the real artifact and how much I improved my hands-on skills through the process, especially if I compare it with the first tryings .

Visuals



Figure 1. Original charger reference.

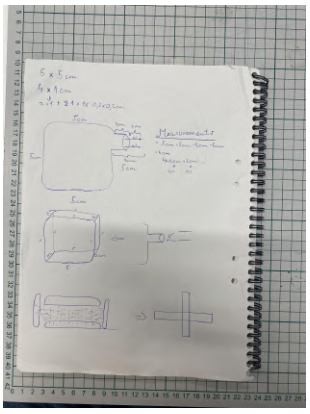


Figure 2. Measurements.

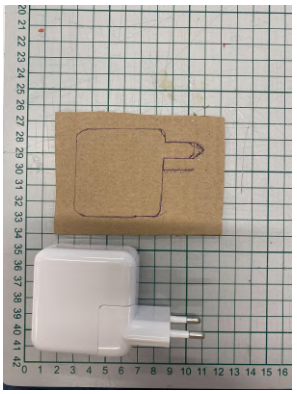


Figure 3. Get to know with carton paper.

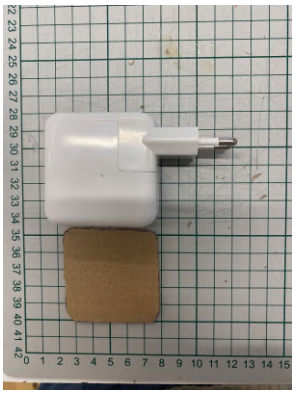


Figure 4. Assembly process.

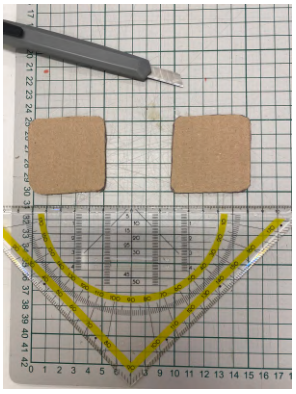


Figure 5. Measuring with the stock.



Figure 6. Cut out to stock(7pcs).

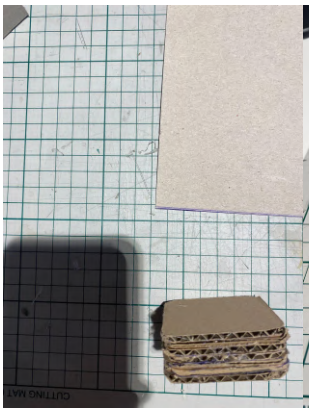


Figure 7. Sticking the stocks together.



Figure 8. Covering the edge of the dock.



Figure 9. Covering the upside, downside.



Figure 10. Creating logo in laser cutter.



Figure 11. Logo.

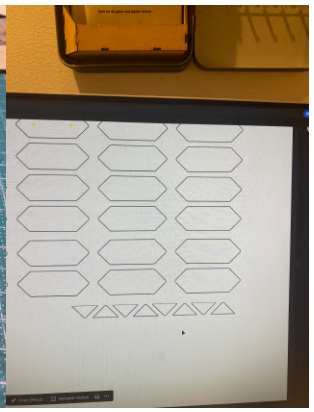


Figure 12. Creating the charger part in laser cutter.

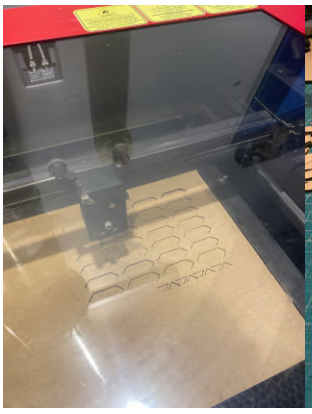


Figure 13. Cutting part of the charger.



Figure 14. Cut down parts and logo on model.



Figure 15. Sticking together two parts.



Figure 16. USB C carving.



Figure 17. USB C fitting in model.



Figure 18. Creating cover for second part.

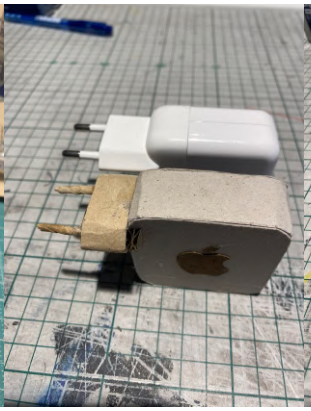


Figure 19. Final product.

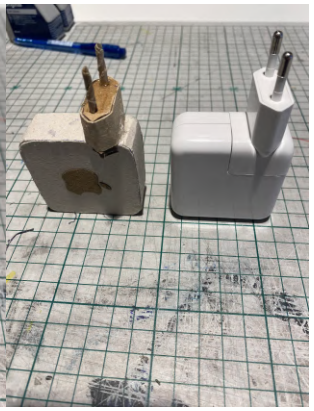


Figure 20. Final product overview.